A Log-Linear Regression Analysis of Allometric Scaling and Species Effects on Penguin Body Mass

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Abstract

This project studies the relationship between penguin body mass and selected morphological traits using the Palmer Penguins dataset. The response variable is body mass, while flipper length, culmen length, culmen depth, and species are used as predictors. Since biological measurements often follow multiplicative scaling relationships, both the response and the continuous predictors are log-transformed, yielding a multiple log-linear regression model.

Species is incorporated through dummy variables, with Adelie taken as the baseline category. The fitted model is estimated using Ordinary Least Squares and explains a substantial proportion of the variation in log body mass, with $R^2 = 0.8353$ and adjusted $R^2 = 0.8328$. All predictors except the intercept are statistically significant at the 5% level.

Confidence intervals, prediction intervals, and hypothesis tests are constructed and interpreted. Residual diagnostics, including plots, a normal Q-Q plot, and a chi-square goodness-of-fit test, suggest that the assumptions of the log-linear model are reasonably satisfied. The analysis shows that penguin body mass is strongly associated with morphological traits and species membership.

Contents

1 Introduction

Understanding how body mass relates to physical characteristics is a central problem in biological data analysis. Structural measurements such as flipper length and bill dimensions are often closely linked to overall size. Studying these relationships helps in both prediction and in understanding biological variation across species.

In this project, we analyze the Palmer Penguins dataset, which contains measurements of penguins from the Palmer Archipelago, Antarctica. The objective is to model body mass using flipper length, culmen length, culmen depth, and species. Since both continuous and categorical predictors are present, multiple linear regression provides a natural framework.

Biological traits frequently exhibit multiplicative scaling, rather than purely additive relationships. To account for this, a log transformation is applied to the response and the continuous predictors. This leads to a log-linear model that is both statistically convenient and biologically motivated through allometry.

The analysis begins with a description of the data and variables. A regression model is then specified using log-transformed predictors and dummy variables for species. The model is estimated using Ordinary Least Squares, followed by interpretation of coefficients, construction of confidence and prediction intervals, and hypothesis testing. Finally, residual diagnostics are used to assess the validity of the model assumptions.

2 Data Description

The dataset used in this study is the Palmer Penguins dataset. It contains morphological measurements of penguins observed in the Palmer Archipelago, Antarctica. It includes data for three species: Adelie, Chinstrap, and Gentoo.

- **Body Mass (grams)**: The weight of the penguin, which serves as the response (dependent) variable.
- **Flipper Length (mm)**: The length of the penguin's flipper, used as a predictor.
- **Culmen Length (mm)**: The length of the penguin's bill (beak), measured along the top ridge, used as a predictor.
- **Culmen Depth (mm)**: The depth (height) of the penguin's bill, used as a predictor.
- **Species**: A categorical variable indicating the species of the penguin (Adelie, Chinstrap, or Gentoo), also used as a predictor.

The dataset initially contained a small number of missing values in some of the numerical variables. These observations were removed prior to analysis to ensure consistency and reliability of the regression models.

The variables selected for this study are biologically relevant, as body mass is expected to be influenced by structural features such as flipper size and bill dimensions, along with species-specific characteristics.

2.1 Summary Statistics

Variable	Mean	Minimum	Maximum	Std. Dev.
Body Mass (g)	4201.754	2700	6300	801.9545
Flipper Length (mm)	200.9152	172	231	14.06171
Culmen Length (mm)	43.92193	32.1	59.6	5.459584
Culmen Depth (mm)	17.15117	13.1	21.5	1.974793

Table 1: Overall Summary Statistics

Species	Variable	Mean	Minimum	Maximum	Std. Dev.
Adelie (151)	**Body Mass (g)**	3700.662	2850	4775	458.5661
	Flipper Length (mm)	189.9536	172	210	6.539457
	Culmen Length (mm)	38.79139	32.1	46	2.663405
	Culmen Depth (mm)	18.34636	15.5	21.5	1.21665
Gentoo (123)	**Body Mass (g)**	5076.016	3950	6300	504.1162
	Flipper Length (mm)	217.187	203	231	6.484976
	Culmen Length (mm)	47.50488	40.9	59.6	3.081857
	Culmen Depth (mm)	14.98211	13.1	17.3	0.9812198
Chinstrap (68)	**Body Mass (g)**	3733.088	2700	4800	384.3351
	Flipper Length (mm)	195.8235	178	212	7.131894
	Culmen Length (mm)	48.83382	40.9	58	3.339256
	Culmen Depth (mm)	18.42059	16.4	20.8	1.135395

Table 2: Summary Statistics of Species

2.2 Data Source

The dataset used in this study is the Palmer Penguins[1] dataset, originally provided by Dr. Kristen Gorman. It is available via the GitHub repository: palmerpenguins (GitHub), and is also hosted on Kaggle: Palmer Archipelago Penguin Data (Kaggle).

3 Model Specification

3.1 Theoretical Motivation

Biological traits (such as beak dimensions, flipper length, and body mass) scale proportionally relative to one another, a phenomenon known as allometry. This proportional, multiplicative growth is mathematically represented by a non-linear power law equation: $Y = \alpha X^{\beta}$.

Ordinary Least Squares (OLS) regression can only fit strictly linear relationships. By applying a natural logarithm to both sides of the allometric power law, we algebraically transform the biological curve into a perfectly straight line:

$$\log(Y) = \log(\alpha) + \beta \log(X).$$

This mathematical linearization allows the standard OLS model to accurately capture and estimate the proportional scaling rate (β) as a simple additive slope.

3.2 Model Structure and Variable Encoding

First, we briefly introduce categorical variables before putting them to use. Unlike continuous variables (such as Flipper Length or Body Mass) which represent measurable, numerical quantities, categorical variables represent qualitative groups or labels (such as Penguin Species). Ordinary Least Squares (OLS) regression relies on strictly numerical equations. Therefore, categorical variables must be mathematically transformed before they can be included in a linear model.

To incorporate qualitative data, statistical models utilize Dummy Variables (also known as One-Hot Encoding). This process translates categorical labels into a series of binary variables, assigning a 1 if the condition is met and a 0 if it is not.

In context of our current dataset, we are going to use the following categorical variables:

- `Is_Gen`: 1 if Gentoo, 0 otherwise.
- `Is_Chin`: 1 if Chinstrap, 0 otherwise.

The baseline, Adelie, is mathematically implied when both `Is_Gen` and `Is_Chin` are equal to 0. We do not account for this separately in the model as its effect is absorbed directly into the model's intercept, β_0.

Our objective is to predict the response variable, `Body_M`, using the predictors `Flip_L`, Culm_L, Culm_D, Is_Gen, and Is_Chin. We have $n = 342$ row entries in our dataset and $p = 5$ many parameters. Clearly $n > p + 1$. Furthermore, if we let X be the $n \times (p + 1)$ order matrix where the first column has all entries 1 and the other columns have row-wise entries of the values of the predictors stated above, in order, then X has full column rank. Thus, the model is estimable and does not suffer from perfect multicollinearity. The remaining regression assumptions are assessed through residual diagnostics.

We shall now verify the credibility of the normality assumption of the errors. First, we fit a model using the raw data from our dataset and then compare it with the model fitted by using the log-transformed data.

We know that $\hat{\beta}$ is given by $(X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}Y$. Plugging in the values for the raw data, we get that

$$\hat{\beta}_{\text{raw}} = (-4327.327, 20.241, 41.468, 140.328, 934.887, -513.247)^{\mathsf{T}}$$

for which the corresponding fitted model is

$$\begin{aligned}\widehat{\texttt{Body\_M}} = -4327.327 + 20.241 \cdot \texttt{Flip\_L} + 41.468 \cdot \texttt{Culm\_L} + 140.328 \cdot \texttt{Culm\_D} \\ + 934.887 \cdot \texttt{Is\_Gen} - 513.247 \cdot \texttt{Is\_Chin}.\end{aligned}$$

So, the residuals, $\hat{\varepsilon}_i$ are given by $\texttt{Body\_M}_i - \widehat{\texttt{Body\_M}}_i$. Two residual plots for the raw model are shown below: a plot of residuals against observation index, and a histogram of the residuals.

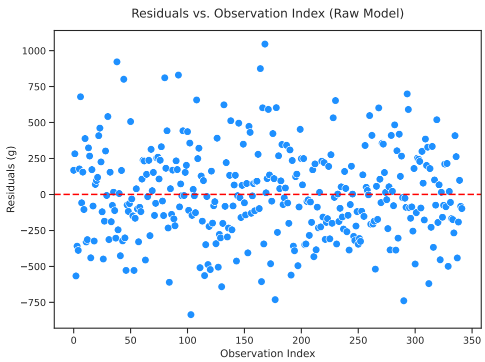

Figure 1: Plot of Residuals vs. Observation Index for the Raw Model

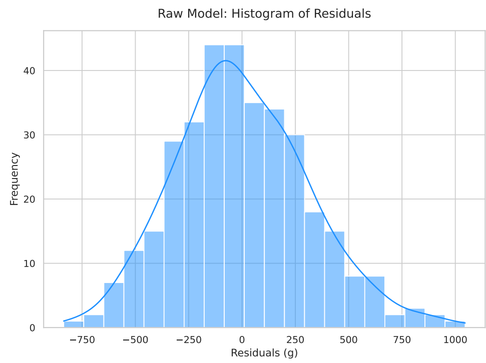

Figure 2: Histogram of Residuals for the Raw Model

As it can be noted, in the histogram plot of the residuals (Fig. 2), the distribution is skewed towards the right. Thus, this might pose problems for the normality assumption.

Now, let us work with the log-transformed model. Proceeding similarly as before, we get

$$\hat{\beta}_{\log} = (0.07590, 0.90358, 0.43644, 0.61806, 0.23408, -0.11952)^{\mathsf{T}}$$

for which the corresponding fitted model is

$$\widehat{\log(\texttt{Body\_M})} = 0.07590 + 0.90358 \cdot \log(\texttt{Flip\_L}) + 0.43644 \cdot \log(\texttt{Culm\_L}) \\ + 0.61806 \cdot \log(\texttt{Culm\_D}) + 0.23408 \cdot \texttt{Is\_Gen} - 0.11952 \cdot \texttt{Is\_Chin}.$$

In this case, the residuals, $\hat{\varepsilon}_i$ are given by $\log(\texttt{Body\_M}_i) - \widehat{\log(\texttt{Body\_M}_i)}$. Two residual plots for the log-transformed model are shown below: a plot of residuals against observation index, and a histogram of the residuals.

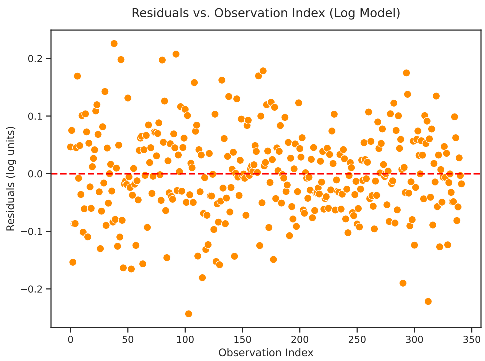

Figure 3: Plot of Residuals vs. Observation Index for Log Model

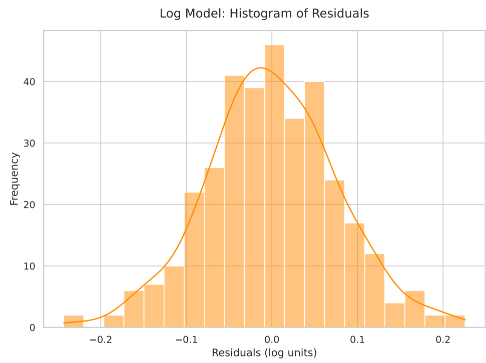

Figure 4: Histogram of Residuals for the Log Model

In the histogram of the log-transformed model (Fig. 4), the curve looks almost perfectly symmetric and the left and right tails mirror each other around the center zero mark.

This suggests us that using the log-transformed data would be better suited to our needs. We shall further verify the goodness of this fit test in Section 5.3.

4 Model Estimation and Results

4.1 Estimated Model

The Multiple Linear Regression (MLR) model was fitted to the data using Ordinary Least Squares (OLS) estimation. As established in the previous section, the fitted regression equation utilizes log-transformed continuous variables to linearize the underlying allometric scaling relationships, and is represented as:

$$\widehat{\log(\texttt{Body\_M})} = 0.07590 + 0.90358 \cdot \log(\texttt{Flip\_L}) + 0.43644 \cdot \log(\texttt{Culm\_L}) \\ + 0.61806 \cdot \log(\texttt{Culm\_D}) + 0.23408 \cdot \texttt{Is\_Gen} - 0.11952 \cdot \texttt{Is\_Chin}.$$

The following scatterplot matrix shows the pairwise relationship between our different variables used for this study:

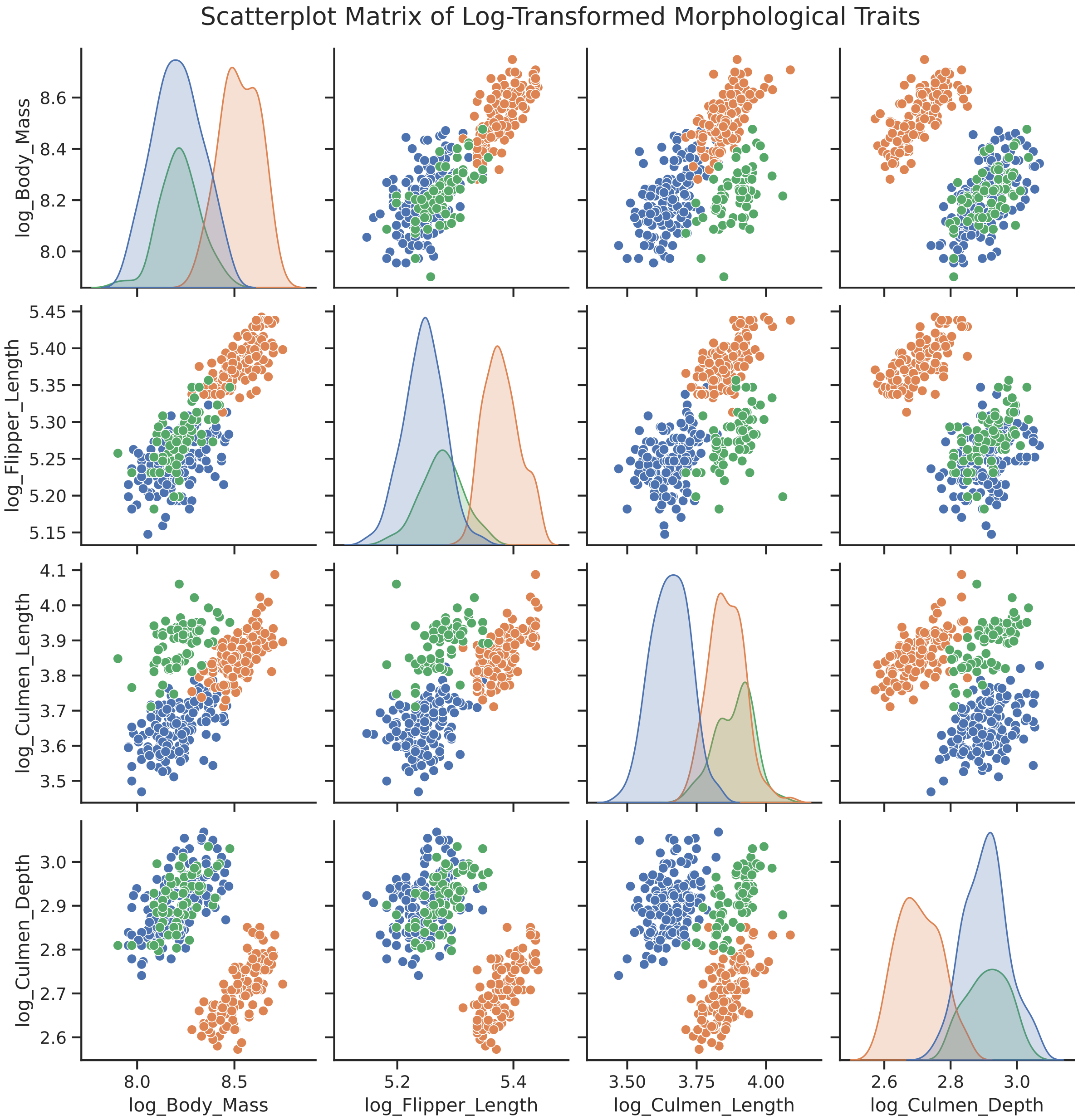

Figure 5: Scatterplot matrix of log-transformed morphological traits. Data points and density distributions are colored by species: Adelie (blue), Gentoo (orange), and Chinstrap (green).

4.2 Regression Results

The results of the regression analysis, including the estimated coefficients, standard errors, and corresponding p-values, are summarized in Table 3. The model is highly robust, explaining approximately 83.5% of the variance in the log-transformed body mass of the 342 penguins sampled ($R^2 = 0.8353$).

Predictor Variable	Coefficient ($\hat{\beta}$)	Std. Error	p-value
`Intercept` ($\hat{\beta}_0$)	0.07590	0.677	0.911
log(`Flip_L`) ($\hat{\beta}_1$)	0.90358	0.149	< 0.001
log(`Culm_L`) ($\hat{\beta}_2$)	0.43644	0.076	< 0.001
log(`Culm_D`) ($\hat{\beta}_3$)	0.61806	0.080	< 0.001
`Is_Gen` ($\hat{\beta}_4$)	0.23408	0.035	< 0.001
`Is_Chin` ($\hat{\beta}_5$)	-0.11952	0.020	< 0.001

Table 3: OLS Regression Results for Penguin Body Mass

The following correlation matrix shows the correlation between the variables used in this study:

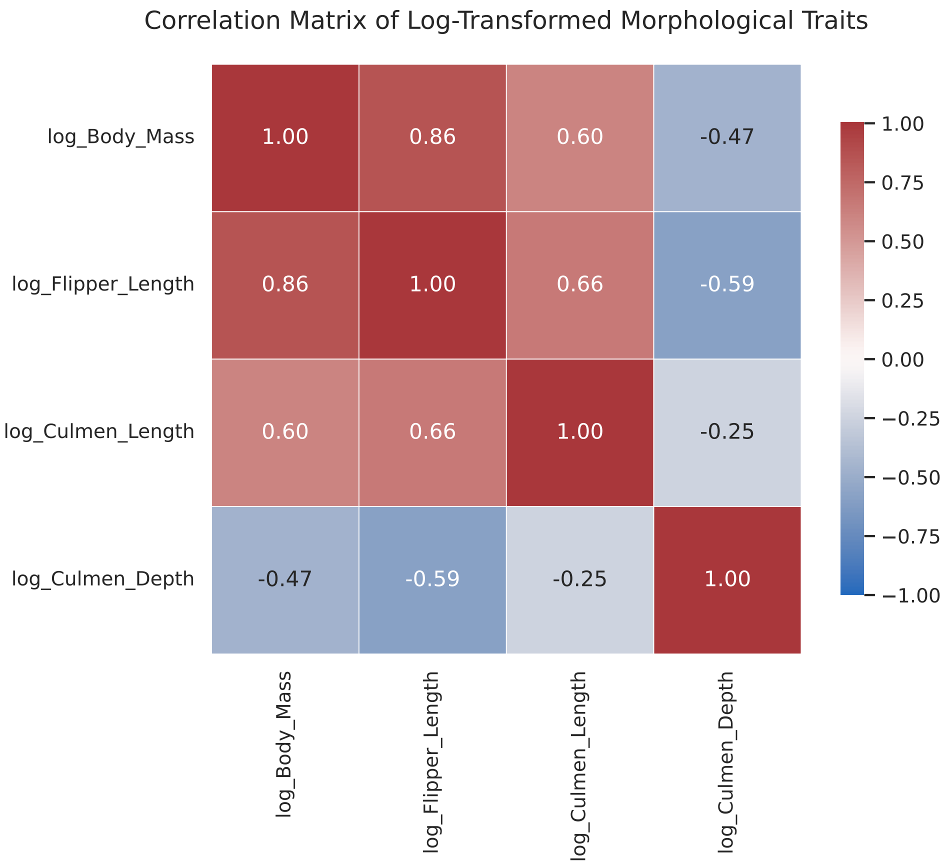

Figure 6: Correlation Matrix of Log-Transformed Morphological Traits

The correlation matrix of the log-transformed variables reveals clear linear relationships among the morphological traits. In particular, log(`Body_M`) shows a strong positive correlation with log(`Flipper_L`), indicating that larger penguins tend to have longer flippers. There is also a moderate positive association between log(`Body_M`) and log(`Culmen_L`), suggesting that heavier penguins generally have longer bills. In contrast, log(`Culmen_D`) exhibits a strong negative relationship with log(`Body_M`) and other variables, suggesting a different structural relationship across species.

4.3 Interpretation of Coefficients

The regression results provide highly significant insights into the physical and species-related differences associated with penguin body mass. Notably, every predictor variable (excluding the baseline intercept) proved to be statistically significant at the $\alpha = 0.05$ level:

- **Allometric Scaling:** All physical dimensions exhibit a positive and highly significant relationship with log(`Body_M`) ($p < 0.001$).
  - For every 1% increase in Flipper Length (`Flip_L`), body mass (`Body_M`) increases by roughly 0.90%.
  - For every 1% increase in Culmen Length (`Culmen_L`), body mass (`Body_M`) increases by roughly 0.44%.
  - For every 1% increase in Culmen Depth (`Culmen_D`), body mass (`Body_M`) increases by roughly 0.62%.
- **Species Genetics:** The inclusion of the dummy variables successfully controls for underlying genetic differences between the species, with Adelie acting as the unstated baseline reference group. Both indicator variables are highly significant ($p < 0.001$).
  - The positive coefficient for `Is_Gen` (0.23408) indicates that, assuming identical physical proportions, a Gentoo penguin is expected to have approximately 26.4% greater body mass than an Adelie penguin, holding other variables constant.
  - Conversely, the negative coefficient for `Is_Chin` (-0.11952) dictates that, assuming identical physical proportions, a Chinstrap penguin is expected to have approximately -11.3% lower body mass than an Adelie penguin, holding other variables constant.

5 Statistical Inference

5.1 Confidence Intervals

In multiple linear regression, estimating the point values for the coefficients ($\hat{\beta}_i$) provides a single best guess for the population parameters. However, to account for sampling variability and quantify the precision of these estimates, we construct 95% confidence intervals for each predictor, the mean response, and future individual observations.

5.1.1 Coefficient Intervals

The true parameter ($\hat{\beta}_i$) is estimated to fall within the following interval with a 95% level of confidence:

$$\hat{\beta}_i \pm t_{\alpha/2}(n-p-1) \cdot SE(\hat{\beta}_i).$$

Plugging in the values, we derive the following interval estimates:

Predictor Variable	**Estimate**	**Lower Bound** (2.5%)	**Upper Bound** (97.5%)
`Intercept` ($\hat{\beta}_0$)	0.0759	-1.2559	1.4077
log(`Flipper_Length`) ($\hat{\beta}_1$)	0.9036	0.6099	1.1972
log(`Culmen_Length`) ($\hat{\beta}_2$)	0.4364	0.2864	0.5865
log(`Culmen_Depth`) ($\hat{\beta}_3$)	0.6181	0.4598	0.7763
`Is_Gen` ($\hat{\beta}_4$)	0.2341	0.1648	0.3034
`Is_Chin` ($\hat{\beta}_5$)	-0.1195	-0.1592	-0.0798

Table 4: 95% Confidence Intervals for the Regression Parameters

Because both the response variable and the continuous predictors were log-transformed, the coefficients are interpreted as elasticities, representing the expected percentage change in body mass for a 1% increase in the predictor, holding all other variables constant.

- **Flipper Length:** We are 95% confident that for every 1% increase in a penguin's flipper length, its expected body mass increases by between 0.61% and 1.20%.
- **Culmen Length:** We are 95% confident that a 1% increase in culmen length corresponds to an expected body mass increase of between 0.29% and 0.59%.
- **Culmen Depth:** We are 95% confident that a 1% increase in culmen depth is associated with a 0.46% to 0.78% increase in expected body mass.
- **Species Effects (`Is_Gen`):** We are 95% confident that, holding all physical measurements constant, a Gentoo penguin will weigh between 17.9% and 35.4% more

than the baseline Adelie species.

- **Species Effects (`Is_Chin`):** We are 95% confident that, holding all physical measurements constant, a Chinstrap penguin will weigh between 7.7% and 14.7% less than the Adelie species.
- **Intercept:** The 95% confidence interval for the intercept ranges from -1.2559 to 1.4077. The intercept corresponds to the expected log body mass of a baseline Adelie penguin when all log-transformed predictors are equal to zero. This implies that the original measurements are equal to 1 (since $\log(1) = 0$), which is not biologically meaningful. Thus, the intercept has limited practical interpretation in this context.

5.1.2 Confidence Intervals for the Mean Response

Let x_0 be a column vector containing a specific set of predictor values. To estimate the expected value $\mathbb{E}[\log(\texttt{Body\_M})]$ at x_0, we utilize the matrix X to account for the covariance among predictors. The 95% confidence interval for the mean response is given by:

$$\hat{y}_0 \pm t_{\alpha/2}(n-p-1) \cdot \sqrt{\text{MSE} \cdot x_0^T (X^T X)^{-1} x_0}.$$

Here, MSE is the Mean Squared Error, estimating the error variance σ^2.

Application: Expected Mass Across Species To isolate the structural effects of the categorical variables, we construct a theoretical standardized penguin utilizing the mean of the log-transformed morphological measurements. We then predict the expected $\log(\texttt{Body\_M})$ across all three species profiles, keeping these baseline log-dimensions mathematically constant.

Species	Point Estimate	Lower Bound	Upper Bound
Adelie (Baseline)	8.2651	8.2337	8.2964
Chinstrap	8.1456	8.1178	8.1733
Gentoo	8.4992	8.4581	8.5402

Table 5: 95% Confidence Intervals for Mean Response

Observing the Chinstrap and Gentoo confidence bounds, we see parallel shifts relative to the Adelie baseline. Because the input traits are identical, we are 95% confident that the average log body mass of a Gentoo population will always remain strictly higher than an identically proportioned Adelie population, whereas a Chinstrap population will remain strictly lower.

5.1.3 Prediction Intervals

While the confidence interval predicts $\mathbb{E}[\log(\texttt{Body\_M})]$ for all penguins with a specific set of traits, the prediction interval predicts the exact $\log(\texttt{Body\_M})$ of a single, individual new penguin. Because individual observations naturally vary around the population mean, predicting a single point carries additional irreducible uncertainty. The 95% prediction interval of $\log(\texttt{Body\_M})$ is mathematically wider and is expressed as:

$$\hat{y}_0 \pm t_{\alpha/2}(n-p-1) \cdot \sqrt{\text{MSE} \cdot (1 + x_0^T (X^T X)^{-1} x_0)}.$$

The addition of 1 inside the parenthetical term accounts for the variance of the residual error (σ^2) associated with drawing a new, random individual from the population.

Application: Predicting Individual Penguins Using the same theoretical baseline log-dimensions, we now calculate the intervals for randomly sampling a single new penguin from each respective species.

Species	Point Estimate	Lower Bound	Upper Bound
Adelie (Baseline)	8.2651	8.1107	8.4195
Chinstrap	8.1456	7.9918	8.2993
Gentoo	8.4992	8.3425	8.6558

Table 6: 95% Prediction Intervals for New Observations

As mathematically expected, these prediction intervals are significantly wider than their corresponding confidence intervals. This increased width explicitly accounts for the inherent random variation (σ^2) between individual penguins, even when they belong to the exact same species and share identical physical measurements.

5.2 Hypothesis Testing

To formally evaluate the explanatory power and structural validity of our multiple log-linear regression model, we conduct three distinct tiers of hypothesis testing.

5.2.1 Overall Model Significance

The global F-test evaluates whether the model as a whole explains a statistically significant portion of the variance in expected log body mass. To do this, we compare the full model against a restricted, intercept-only baseline model.

- **Null Hypothesis (H_0):** $\beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$. None of the predictors have a significant linear relationship with the response.
- **Alternative Hypothesis (H_1):** At least one $\beta_i \neq 0$ (for $i \in \{1, 2, 3, 4, 5\}$).

The test statistic evaluates the reduction in SSE when all predictors are added to the intercept model:

$$F_0 = \frac{(\text{SSE}_{\text{restricted}} - \text{SSE}_{\text{full}})/\text{df}_{\text{restricted}} - \text{df}_{\text{full}}}{\text{SSE}_{\text{full}}/\text{df}_{\text{full}}}.$$

For the global test, the restricted intercept-only model has $\text{df} = n - 1$, and the full model containing 5 predictors has $\text{df} = n - 6$. Evaluating these degrees of freedom, the formula simplifies to:

$$F_0 = \frac{(\text{SSE}_{\text{restricted}} - \text{SSE}_{\text{full}})/5}{\text{SSE}_{\text{full}}/336}.$$

Calculation and Inference Evaluating the global model yields an F-statistic of $F_0 = 340.7279$, which corresponds to a p-value strictly less than 0.001. Because this p-value is significantly smaller than our significance level ($\alpha = 0.05$), we reject the null hypothesis. We conclude with 95% confidence that the regression model as a whole explains a statistically significant proportion of the variance in expected log body mass.

5.2.2 Individual Parameter Significance

While the global F-test evaluates the entire model, marginal t-tests evaluate the independent contribution of each specific predictor, holding all other variables constant.

- **Null Hypothesis (H_0):** $\beta_i = 0$. The i-th predictor does not significantly affect expected log body mass.
- **Alternative Hypothesis (H_1):** $\beta_i \neq 0$.

The test statistic is calculated by dividing the parameter estimate by its standard error:

$$t_0 = \frac{\hat{\beta}_i}{\text{SE}(\hat{\beta}_i)}.$$

Predictor Variable	Estimate ($\hat{\beta}_i$)	t-statistic	p-value
`Intercept`	0.0759	0.112	0.911
log(`Flipper_Length`)	0.9036	6.053	< 0.001
log(`Culmen_Length`)	0.4364	5.722	< 0.001
log(`Culmen_Depth`)	0.6181	7.682	< 0.001
`Is_Gen`	0.2341	6.641	< 0.001
`Is_Chin`	-0.1195	-5.925	< 0.001

Table 7: Marginal t-Tests for Individual Parameters

Inference As shown in the table, with the exception of the intercept, every morphological and categorical predictor yields a p-value strictly less than 0.001. This strongly rejects the null hypotheses at the $\alpha = 0.05$ level, confirming that each variable is a highly significant independent predictor of log body mass.

5.2.3 Joint Significance of Categorical Variables

To rigorously justify the inclusion of the categorical species indicators, we test their joint significance against a restricted model containing only the physical measurements.

- **Null Hypothesis (H_0):** $\beta_4 = \beta_5 = 0$. The species classifications provide no additional explanatory power beyond physical dimensions.
- **Alternative Hypothesis (H_1):** At least one species parameter is non-zero.

We use the same F-statistic as the global test:

$$F_0 = \frac{(\text{SSE}_{\text{restricted}} - \text{SSE}_{\text{full}})/\text{df}_{\text{restricted}} - \text{df}_{\text{full}}}{\text{SSE}_{\text{full}}/\text{df}_{\text{full}}}.$$

For our specific partial test, the restricted model contains 3 continuous predictors (df $= n - 4$), and the full model adds 2 categorical indicators (df $= n - 6$). Evaluating the degrees of freedom yields:

$$F_0 = \frac{(\text{SSE}_{\text{restricted}} - \text{SSE}_{\text{full}})/2}{\text{SSE}_{\text{full}}/336}.$$

Calculation and Inference Evaluating the joint addition of the species variables yields an F-statistic of $F_0 = 95.7775$ with a corresponding p-value strictly less than 0.001. Because this p-value is significantly smaller than $\alpha = 0.05$, we firmly reject the null hypothesis. This mathematically suggests that the categorical species indicators provide a highly significant improvement to the model's predictive power, reasonably justifying their inclusion over a model based purely on physical dimensions.

5.3 Model Diagnostics

To conclude our analysis, we must verify both the explanatory power of the model and the structural validity of the Ordinary Least Squares (OLS) assumptions in log-space.

5.3.1 Goodness of Fit

The explanatory power of the model is measured by the proportion of total variance in log(`Body_M`) that is captured by our selected predictors.

- **Multiple R^2 (0.8353):** Our model explains 83.53% of the sample variance in the observed log(`Body_M`).
- **Adjusted R^2 (0.8328):** Because adding parameters mathematically inflates the raw R^2. Our high adjusted R^2 suggests that the inclusion of the categorical species indicators provides a genuine increase in predictive accuracy.

5.3.2 Residual Analysis

A regression model is only mathematically valid if its unobserved error terms (ε) are random and independent.

- **Residuals vs. Fitted:** The plotted residuals display a random scatter clustered tightly around the horizontal zero-line with no distinct funnels or curves. This suggests that the variance of the errors is constant and that the log-linear structural relationship is appears appropriate.

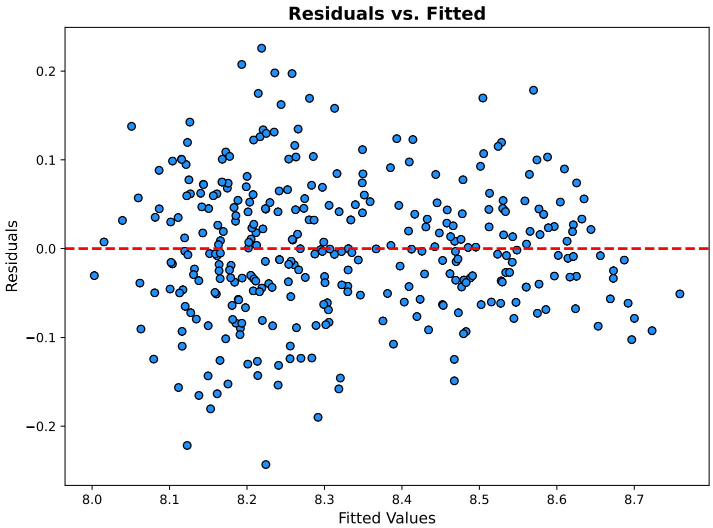

Figure 7: Residuals vs. Fitted

- **Normal Q-Q Plot:** The standardized residuals tightly hug the theoretical 45-degree diagonal line, visually indicating that the error terms are normally distributed.

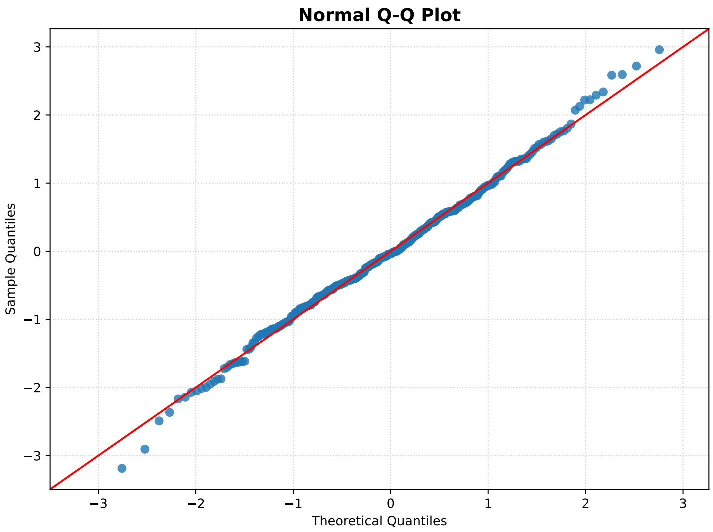

Figure 8: Normal Q-Q Plot

5.3.3 Normality Assessment

To rigorously prove the visual assumption of normality established by the Q-Q plot, we apply the Pearson Chi-Square Goodness-of-Fit test to the model's standardized residuals.

By standardizing the residuals $(\frac{\varepsilon-\bar{\varepsilon}}{s})$, we map them directly to a standard Normal distribution curve, $N(0, 1)$. We slice this theoretical curve into $k = 10$ regions with equal probabilities, meaning the expected probability of a residual falling into any specific region is exactly $E_i = 10\%$.

We then map our actual observed standardized residuals (O_i) into these regions. The test statistic evaluates the difference between the observed counts and the theoretically expected counts:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}.$$

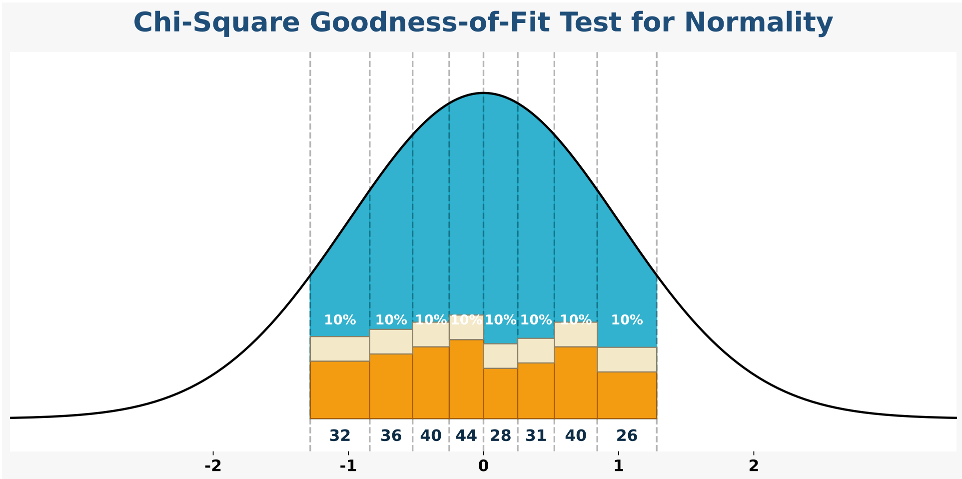

Figure 9: Chi-Square Goodness-of-Fit Test for Normality

Calculation and Inference Evaluating the standardized residuals across 10 equiprobable bins yields a test statistic of $\chi^2 = 9.6374$. With $k - 3 = 7$ degrees of freedom (accounting for the estimation of the mean and variance), this corresponds to a p-value of 0.2101.

Because this p-value is strictly greater than our significance threshold ($\alpha = 0.05$), we **fail to reject** the null hypothesis. This mathematically suggests that our residuals do not significantly deviate from a normal distribution. Consequently, all confidence intervals, prediction intervals, and hypothesis tests calculated in the preceding sections are reasonably supported.

6 Conclusion

This study developed a multiple linear regression model to explain penguin body mass using flipper length, culmen length, culmen depth, and species. Motivated by allometric scaling, a log transformation was applied to obtain a log-linear model that better captures the underlying relationships.

The fitted model demonstrated strong explanatory power, accounting for about 83.5% of the variation in log body mass. All predictors, except the intercept, were highly significant. Flipper length, culmen length, and culmen depth each showed a positive association with body mass, while the species indicators captured meaningful differences among penguin species.

The inferential results supported the usefulness of the model. The overall F-test confirmed statistical significance, and the joint test showed that species provides additional explanatory power beyond physical measurements. Confidence and prediction intervals were consistent with the fitted model.

Diagnostic analysis indicated that the regression assumptions are reasonably satisfied. Residual plots showed no major patterns, the Q-Q plot suggested approximate normality, and the chi-square test did not detect significant deviations. Overall, the model provides both strong predictive performance and a meaningful representation of the relationship between body mass and morphological traits.

A Appendix

A.1 Dataset

The dataset used in this study is the Palmer Penguins dataset[1], originally collected by Kristen B. Gorman and made publicly available by Allison Marie Horst, Alison Presmanes Hill, and Kristen B. Gorman.

It can be accessed through the following sources:

- GitHub: github.com/allisonhorst/palmerpenguins
- Kaggle: kaggle.com/datasets/parulpandey/palmer-archipelago-antarctica-penguin-data

A.2 Code

The code used in this project is divided based on functionality:

- R Code (Computation and Model Fitting): R Code Link
- Python Code (Visualization and Plotting): Python Code Link

References

[1] Allison Marie Horst, Alison Presmanes Hill, and Kristen B Gorman. *palmerpenguins: Palmer Archipelago (Antarctica) penguin data*, 2020. R package version 0.1.0.